

# CFM 301 -- Project 1: Constructing the Most Efficient Portfolio

SAMPLE REPORT -- illustrative only, not a submission. Group [X] · [Names] · [Date]

## Economic Goal & Motivation

Index concentration is a live debate in practice: a handful of mega-cap stocks drive most of a market-cap-weighted index's risk and return, and critics argue this leaves investors under-diversified relative to what they think they own. This project asks a sharper, testable version of that question using the 15 largest U.S. stocks just before the 2008 financial crisis: **does a mean-variance-optimal (minimum-variance) investor want to hold these stocks in roughly the same proportions as the market does, or does the optimizer actively steer away from the biggest, most-correlated names -- and if the two portfolios differ, did that difference matter once the crisis actually hit?**

## Data Description

**Source:** CRSP daily stock file (`crsp.dsf`) and delisting file (`crsp.dsdelist`), accessed via WRDS, linked to Compustat via the CRSP/Compustat Merged (CCM) link table. **Universe:** the 15 largest U.S. common stocks by market capitalization (price × shares outstanding) as of the end of the estimation window, among stocks with a complete 60-month return history. **Sample period:** January 2003 -- December 2007 (60 monthly observations) for estimation, ending immediately before the 2008 financial crisis; realized returns through December 2009 are used only for the out-of-sample crisis check below. **Cleaning:** daily returns are compounded to monthly; delisting returns are merged onto a stock's final trading date before compounding; any daily observation with a null, non-finite, or sub -100% return is dropped prior to aggregation.

## Estimated Means, Standard Deviations, and Correlation Matrix

Expected returns and the covariance matrix are estimated directly from the 60-month sample (15 assets, well below the 60 time observations, so the sample covariance matrix is well-conditioned without any shrinkage or regularization).

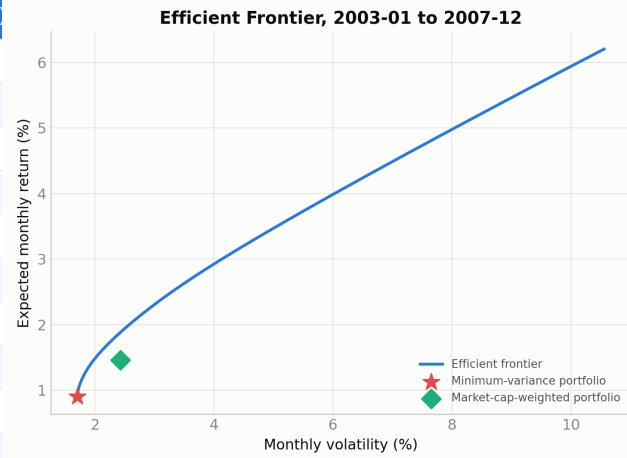
| ID  | Mean (%) | Std (%) | S1    | S2    | S3   | S4    | S5    | S6    | S7    | S8    | S9    | S10   | S11   | S12   | S13   | S14   | S15   |
|-----|----------|---------|-------|-------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| S1  | 1.96     | 5.13    | --    | 0.07  | 0.15 | 0.25  | 0.13  | 0.81  | 0.10  | -0.23 | 0.03  | 0.37  | -0.02 | 0.11  | 0.28  | 0.03  | 0.04  |
| S2  | 1.01     | 4.05    | 0.07  | --    | 0.32 | 0.22  | 0.07  | -0.01 | 0.25  | 0.19  | 0.37  | -0.02 | 0.36  | -0.08 | -0.09 | 0.19  | 0.16  |
| S3  | 0.96     | 5.76    | 0.15  | 0.32  | --   | 0.28  | 0.08  | 0.06  | 0.12  | 0.01  | 0.04  | 0.23  | 0.21  | 0.04  | 0.11  | 0.14  | 0.32  |
| S4  | 1.28     | 5.62    | 0.25  | 0.22  | 0.28 | --    | 0.22  | 0.27  | -0.05 | -0.03 | 0.22  | 0.05  | 0.17  | 0.24  | 0.20  | 0.01  | 0.27  |
| S5  | 1.13     | 3.70    | 0.13  | 0.07  | 0.08 | 0.22  | --    | 0.08  | 0.43  | -0.12 | 0.05  | -0.01 | 0.12  | 0.08  | 0.35  | 0.07  | 0.13  |
| S6  | 2.14     | 5.09    | 0.81  | -0.01 | 0.06 | 0.27  | 0.08  | --    | -0.02 | -0.31 | 0.00  | 0.41  | 0.02  | 0.32  | 0.24  | 0.06  | 0.01  |
| S7  | 0.59     | 3.39    | 0.10  | 0.25  | 0.12 | -0.05 | 0.43  | -0.02 | --    | 0.07  | 0.13  | -0.13 | -0.15 | -0.11 | 0.14  | -0.03 | 0.11  |
| S8  | 0.10     | 4.54    | -0.23 | 0.19  | 0.01 | -0.03 | -0.12 | -0.31 | 0.07  | --    | -0.10 | -0.11 | -0.06 | -0.35 | -0.27 | 0.16  | 0.37  |
| S9  | 0.71     | 3.81    | 0.03  | 0.37  | 0.04 | 0.22  | 0.05  | 0.00  | 0.13  | -0.10 | --    | 0.14  | 0.24  | 0.12  | 0.29  | 0.15  | 0.03  |
| S10 | 6.20     | 10.56   | 0.37  | -0.02 | 0.23 | 0.05  | -0.01 | 0.41  | -0.13 | -0.11 | 0.14  | --    | 0.06  | 0.30  | -0.03 | 0.31  | -0.02 |
| S11 | 1.47     | 7.30    | -0.02 | 0.36  | 0.21 | 0.17  | 0.12  | 0.02  | -0.15 | -0.06 | 0.24  | 0.06  | --    | 0.07  | 0.20  | 0.53  | 0.10  |
| S12 | 2.17     | 7.40    | 0.11  | -0.08 | 0.04 | 0.24  | 0.08  | 0.32  | -0.11 | -0.35 | 0.12  | 0.30  | 0.07  | --    | 0.14  | 0.11  | 0.24  |
| S13 | -0.10    | 5.40    | 0.28  | -0.09 | 0.11 | 0.20  | 0.35  | 0.24  | 0.14  | -0.27 | 0.29  | -0.03 | 0.20  | 0.14  | --    | 0.03  | -0.00 |
| S14 | 1.31     | 7.94    | 0.03  | 0.19  | 0.14 | 0.01  | 0.07  | 0.06  | -0.03 | 0.16  | 0.15  | 0.31  | 0.53  | 0.11  | 0.03  | --    | 0.10  |
| S15 | 1.19     | 3.87    | 0.04  | 0.16  | 0.32 | 0.27  | 0.13  | 0.01  | 0.11  | 0.37  | 0.03  | -0.02 | 0.10  | 0.24  | -0.00 | 0.10  | --    |

Monthly mean, standard deviation, and full pairwise correlation matrix for all 15 universe stocks. IDs map to CUSIP\_PERMNO security identifiers in the accompanying notebook.

## Selection Criterion, Portfolio Weights, and the Market-Cap Comparison

**Criterion: global minimum variance.** We select the point on the efficient frontier with the lowest attainable portfolio variance, rather than a target expected return or a risk-aversion-weighted utility maximum, since it requires no additional parameter that would itself need separate justification. **Constraints:** long-only and fully invested (weights sum to 1), matching how long-only institutional mandates are actually run. We compare these weights directly against each stock's **market-cap weight** within the same 15-stock group -- the question this project asks.

| ID  | Cap wt (%) | MV wt (%) | Diff (pp) |
|-----|------------|-----------|-----------|
| S1  | 15.0       | 1.2       | -13.8     |
| S2  | 10.9       | 0.0       | -10.9     |
| S3  | 9.7        | 4.2       | -5.5      |
| S4  | 7.4        | 0.0       | -7.4      |
| S5  | 6.7        | 12.4      | +5.7      |
| S6  | 5.8        | 12.7      | +6.9      |
| S7  | 5.6        | 16.2      | +10.6     |
| S8  | 5.6        | 25.1      | +19.6     |
| S9  | 5.4        | 16.5      | +11.1     |
| S10 | 5.1        | 0.0       | -5.1      |
| S11 | 4.8        | 3.0       | -1.8      |
| S12 | 4.7        | 6.6       | +1.9      |
| S13 | 4.5        | 2.1       | -2.5      |
| S14 | 4.5        | 0.0       | -4.5      |
| S15 | 4.5        | 0.0       | -4.5      |



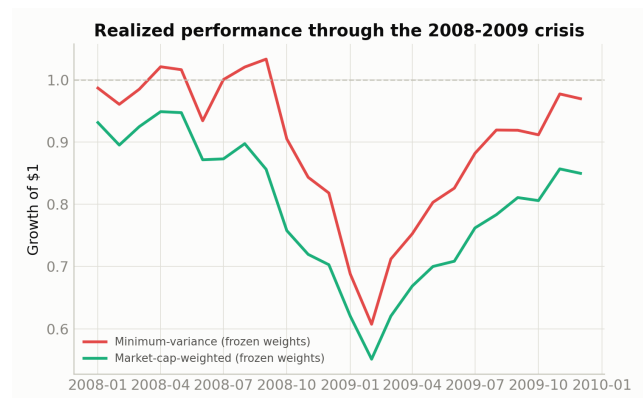
Cap weight vs. minimum-variance weight per stock (ordered by cap weight; IDs match the table above), and the resulting efficient frontier with both portfolios marked.

The minimum-variance portfolio diverges sharply from cap weighting: the largest cap-weighted stock (S1, 15.0% of cap weight) receives only 1.2% minimum-variance weight, while 5 of the 15 stocks receive exactly zero weight despite each being 4-11% of the cap-weighted group. The minimum-variance portfolio has expected monthly return 0.90% and volatility 1.70%, versus 1.46% / 2.43% for the cap-weighted portfolio -- lower expected return, but meaningfully lower risk, exactly as the minimum-variance criterion is designed to deliver. Weights sum to 1.0000 and all are non-negative, satisfying the stated constraints.

## Economic Conclusion

**Did it matter?** Freezing both weight sets at end-2007 and holding them (rebalanced monthly to target) through the 2008-2009 crisis: the minimum-variance portfolio lost **-3.0%** cumulatively, versus **-15.0%** for the cap-weighted portfolio -- roughly 4.9x the drawdown. The answer to this project's economic question is a clear yes on both counts: even among just the 15 largest U.S. stocks, mean-variance optimization produces a portfolio with substantially different composition from market-cap weighting, and that difference translated into a materially better outcome through the one major stress period this sample contains. The result is a caution against treating 'large and liquid' as a synonym for 'already diversified': correlation structure among even the biggest stocks leaves real, exploitable room for risk reduction.

*Growth of \$1, frozen weights (rebalanced monthly to target), January 2008 -- December 2009.*



*Note: this project was also extended to study how portfolio optimality decays over time generally (rolling re-estimation across 1985-2024, a different and larger universe). That extension is not part of this report or its grading scope; see EXTENSION\_not\_graded/ in the submission for the full writeup, code, and figures.*